

Lesson Cards

One accessible curriculum-review card per lesson, built from current app data plus inventory notes and rubric scores.

What Is an LLM?

A learned next-token prediction system

Stage: architecture

Priority: low

Rubric avg: 4.4/5

DEFINITION

A large language model is a learned system that predicts likely next tokens from the context it can see.

METAPHOR

A very large autocomplete engine with learned structure, not a mind or a database.

PROMPT/RESPONSE NOTE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

CHECKPOINT

Which description is most accurate? Correct: A learned prediction system. Incorrect: A conscious reader; A hand-written rulebook.

ILLUSTRATION NEEDED

One-page prompt lifecycle: prompt enters, fixed weights process, next token sampled, response grows.

CURRENT EXPLANATION

Where: It sits between the current input context and the next response token. **Why:** This definition lowers the mystery: the model is powerful because training shaped many weights, not because it has a hidden mind.

BRAIN METAPHOR

Like a brain, many small signals can combine into useful patterns.

VISUAL AID

llm-overview

FEEDBACK

LLMs use learned weights to predict and sample tokens. That is powerful, but it is not consciousness. Choice notes: A conscious reader: Not quite. An LLM processes patterns and probabilities; it does not have awareness. A hand-written rulebook: Not quite. That describes symbolic or rule-based AI. An LLM is a learned prediction system.

RELATIONSHIP

During inference, the current context enters a forward pass, one generated response token is chosen, and that token is appended before the next pass.

BRAIN LIMIT

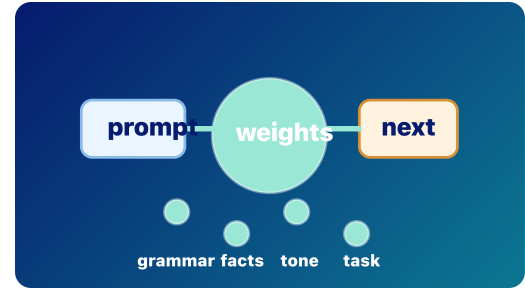
Unlike a brain, an LLM has no lived body, feelings, personal goals, or awareness.

CURRENT EXERCISE

pick-next-token: Pick the Next Token

REWRITE NOTES

Risk: Next-token prediction can sound trivial unless the role of learned weights and repeated inference is made concrete. **Missing:** Add a concrete one-prompt trace that shows context, weights, logits, sampling, and append-repeat at a glance.



Prompt to Prediction
A prompt enters the current context; learned weights shape next-token probabilities.

Rubric Scores				
Accuracy 5/5	Placement 4/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

prompt

response

token

weight

inference

Two AI Traditions

Rules versus learned patterns

Stage: architecture

Priority: medium

Rubric avg: 3.6/5

DEFINITION

Symbolic AI manipulates explicit rules and symbols; deep learning adjusts many weights until useful patterns are captured.

METAPHOR

A rulebook versus a weather system of learned relationships.

PROMPT/RESPONSE NOTE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

CHECKPOINT

Which phrase best describes the LLM tradition? Correct: A learned prediction machine. Incorrect: A fixed rulebook; A spreadsheet macro.

ILLUSTRATION NEEDED

Split-panel rulebook versus learned landscape, with a small note that systems can combine them.

CURRENT EXPLANATION

Where: LLMs mainly come from the deep-learning tradition, even when they imitate rule-like reasoning in language. **Why:** The distinction explains why LLMs can be flexible and fluent without being hand-programmed rule by rule.

BRAIN METAPHOR

The learned-pattern tradition is loosely like practice shaping future behavior.

VISUAL AID

traditions

FEEDBACK

The model was not hand-coded rule by rule; training shaped its weights from examples.

RELATIONSHIP

A prompt is not matched against one fixed rulebook; it is processed by learned parameters that were shaped across many examples.

BRAIN LIMIT

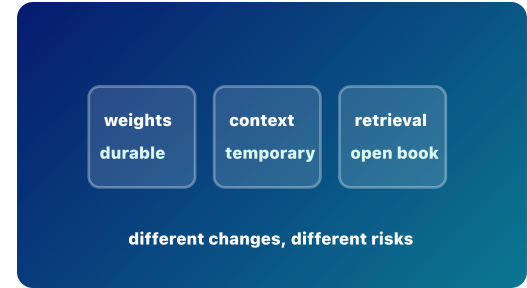
The model is not practicing with intention; training is an optimization process over data.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Learners may read rules versus learned patterns as a total replacement story instead of two still-useful traditions. **Missing:** Name one place rules still matter, such as policies, tool routing, or validation around learned models.



Rules and Learned Patterns

Symbolic AI follows explicit rules; deep learning uses learned parameters.

Rubric Scores

Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 3/5	Exercise 2/5	Mobile 4/5

pretraining training symbolic AI deep learning

Training

Prediction error updates weights

Stage: pretraining

Priority: low

Rubric avg: 4.3/5

DEFINITION

Training uses examples, predicts targets such as next tokens, measures error, and durably updates model weights.

METAPHOR

Practicing a piece until the instrument settings are tuned before the performance.

PROMPT/RESPONSE NOTE

This happens before ordinary prompting; it can change future behavior by updating weights rather than producing one live response.

CHECKPOINT

What makes training different from inference? Correct: Training durably updates weights. Incorrect: Training reads only the context window; Training samples the final token.

ILLUSTRATION NEEDED

Training loop with predict, compare, loss, update weights, repeat.

CURRENT EXPLANATION

Where: It happens before or outside ordinary use, in a separate optimization process. **Why:** Training is where durable capability is installed into the model.

BRAIN METAPHOR

It is fair to say repeated experience shapes later behavior.

VISUAL AID

training-loop

FEEDBACK

The durable update is the key distinction. Inference is a forward pass using existing weights.

RELATIONSHIP

Training changes parameters; inference uses those parameters to generate response tokens without normally rewriting them.

BRAIN LIMIT

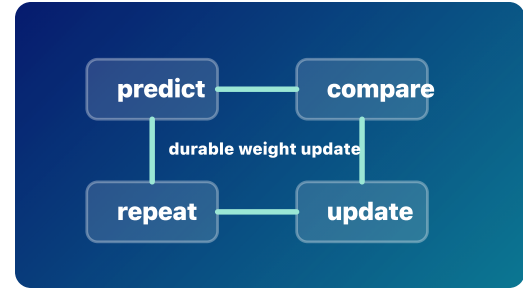
The model has no intention or memory of a lesson; weights are adjusted by math, not reflection.

CURRENT EXERCISE

training-nudge: Training Nudge

REWRITE NOTES

Risk: Ordinary chat can be mistaken for training unless durable weight updates are foregrounded. **Missing:** Show the loss signal and backprop update as the step that ordinary inference lacks.



Training Loop
Predict, compare, update weights, repeat.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

trainingweightparameterinferenceforward pass

Pretraining

Broad durable learning before normal use

Stage: pretraining

Priority: low

Rubric avg: 4.1/5

DEFINITION

Pretraining is broad early training that repeatedly predicts tokens across large datasets and durably updates model weights.

METAPHOR

Billions of tiny nudges carving paths through a vast landscape.

PROMPT/RESPONSE NOTE

This happens before ordinary prompting; it can change future behavior by updating weights rather than producing one live response.

CHECKPOINT

During pretraining, do model weights change? Correct: Yes. Incorrect: No; Only inside one chat.

ILLUSTRATION NEEDED

Broad data rain carving durable pathways into a model landscape.

CURRENT EXPLANATION

Where: It happens before a general model is released for everyday prompting. **Why:** Pretraining gives the model broad language patterns, factual associations, styles, and task structures.

BRAIN METAPHOR

It resembles broad education in the loose sense that later performance depends on prior exposure.

VISUAL AID

pretraining-rain

FEEDBACK

Pretraining is a durable weight-changing process.

RELATIONSHIP

Pretraining changes the model durably; inference only uses the weights that already exist.

BRAIN LIMIT

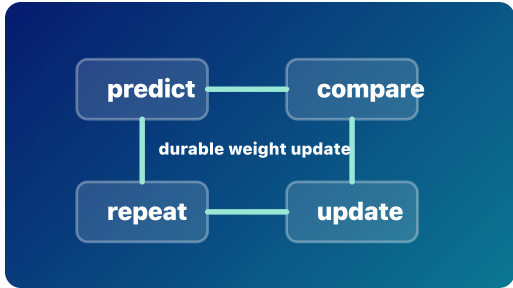
The model does not understand lessons the way a person does; it stores learned statistical structure in parameters.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Broad exposure can be misread as verbatim memorization of every source document. **Missing:** Clarify that broad datasets shape statistical patterns without guaranteeing source recall or truth.



Broad Pretraining

Many prediction updates carve broad capability into model weights.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5

pretraining

training

inference

Fine-Tuning

Targeted durable shaping

Stage: fine-tuning Priority: low Rubric avg: 4.1/5

DEFINITION
Fine-tuning is additional targeted training on examples, preferences, or domain data after broad pretraining.

METAPHOR
Carving useful trails through an already huge terrain.

PROMPT/RESPONSE NOTE
This happens before or between deployments; it changes future response patterns, not the current response token by token.

CHECKPOINT
During ordinary inference, do model weights durably update? Correct: No. Incorrect: Yes; Always after every prompt.

ILLUSTRATION NEEDED
Targeted trail added onto an existing pretrained terrain.

CURRENT EXPLANATION
Where: It happens after pretraining and before or between deployment cycles. **Why:** Fine-tuning can make future responses more helpful, specialized, or aligned with a task.

BRAIN METAPHOR
It is a little like coaching a person toward a style of performance.

VISUAL AID
fine-tune-path

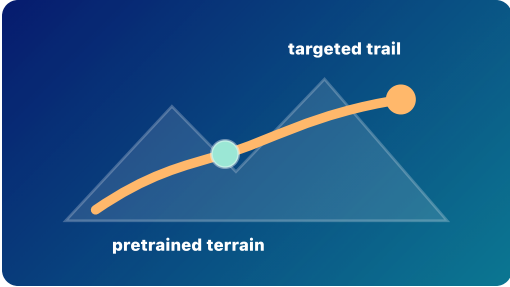
FEEDBACK
Inference computes hidden states and next-token probabilities, but it does not normally change weights.

RELATIONSHIP
Pretraining builds broad capability; fine-tuning can durably change weights or add adapter weights so future responses follow a desired pattern.

BRAIN LIMIT
The model is not adopting values or intentions; the output pattern is being optimized.

CURRENT EXERCISE
None rendered in Journey; no direct Play mapping.

REWRITE NOTES
Risk: Fine-tuning may be confused with prompting, system messages, or one-off personalization. **Missing:** Distinguish full fine-tuning, adapter-style updates, and instruction/policy tuning without adding too much detail.



Fine-Tuning Path
Targeted examples nudge an already-trained model toward a desired pattern.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5

fine-tuning pretraining inference prompt response

Inference

A forward pass, not a memory update

Stage: inference

Priority: low

Rubric avg: 4.4/5

DEFINITION

Inference is normal model use: fixed weights process the current context to generate response tokens without durable weight updates.

METAPHOR

Using a map to choose a route, not redrawing the map.

PROMPT/RESPONSE NOTE

This is the live prompt-to-response path: fixed weights process current context and generate response tokens.

CHECKPOINT

What changes during ordinary inference? Correct: Temporary hidden states. Incorrect: The model weights permanently; The training dataset.

ILLUSTRATION NEEDED

Forward-pass pipeline with fixed weights underneath and temporary states above.

CURRENT EXPLANATION

Where: It happens every time a deployed model answers a prompt. **Why:** This is the line between using a trained model and training a model.

BRAIN METAPHOR

It can feel like thinking because internal representations change temporarily.

VISUAL AID

inference-pass

FEEDBACK

Inference creates temporary internal vectors for this run, then discards or replaces them.

RELATIONSHIP

Each forward pass turns the current context into next-token scores; decoding chooses one generated token for the response.

BRAIN LIMIT

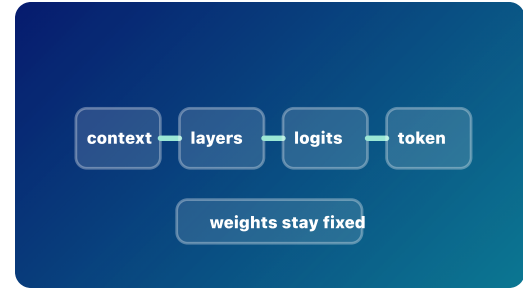
Those hidden states are temporary computations, not memories, beliefs, or learning.

CURRENT EXERCISE

training-nudge: Training Nudge

REWRITE NOTES

Risk: Temporary hidden states may be mistaken for new memories or weight changes. **Missing:** Make the forward pass tangible: fixed weights plus temporary activations produce one set of next-token scores.



Forward Pass

Inference creates temporary states while durable weights stay fixed.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

inference

forward pass

prompt

response

hidden state

logits

Prompt vs Response

Given context versus generated tokens

Stage: prompt processing

Priority: low

Rubric avg: 4.4/5

DEFINITION

The prompt is the input context supplied to the model; the response is the sequence of tokens the model generates after reading it.

METAPHOR

Given cards on a table, then new cards added one at a time.

PROMPT/RESPONSE NOTE

This card should stay explicit that prompt tokens and response-so-far tokens enter the same current context before the next token is chosen.

CHECKPOINT

What happens after one response token is generated? Correct: It is appended to the context. Incorrect: The whole answer appears at once; The model permanently learns it.

ILLUSTRATION NEEDED

Given prompt cards and newly generated response cards entering the next context together.

CURRENT EXPLANATION

Where: Prompt tokens are present before the next token is chosen; response tokens are added one at a time as generation proceeds. **Why:** The distinction prevents a common myth that the model writes a full answer all at once.

BRAIN METAPHOR

It resembles a person continuing a sentence after reading a prompt.

VISUAL AID

prompt–response

FEEDBACK

The response grows by next token, append, repeat.

RELATIONSHIP

Each generated response token is appended to the context, so it can influence the next forward pass.

BRAIN LIMIT

The model is not deciding what it means to say; it samples from learned next-token probabilities.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Learners may think the whole answer appears at once instead of being generated token by token.
Missing: Add the response-so-far as a visible part of the next input context.



Prompt vs Response

Prompt tokens are given; response tokens are generated and appended.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

prompt

response

prompt tokens

response tokens

generated token

context window

Tokenization

Text becomes chunks

Stage: prompt processing

Priority: medium

Rubric avg: 4.2/5

DEFINITION

Tokenization splits prompts and generated responses into chunks the model can represent as token IDs.

METAPHOR

Parcels on a conveyor belt entering the model.

PROMPT/RESPONSE NOTE

This card should stay explicit that prompt tokens and response-so-far tokens enter the same current context before the next token is chosen.

CHECKPOINT

What comes right after text is split into tokens? Correct: Token IDs and embedding lookup. Incorrect: Softmax; A finished answer.

ILLUSTRATION NEEDED

Tokenizer conveyor splitting a short sentence into uneven chunks.

CURRENT EXPLANATION

Where: It happens before embedding lookup and before the transformer layers process the input. **Why:** Models do not carry raw English sentences through their layers; they carry token IDs and numerical vectors.

BRAIN METAPHOR

It loosely resembles hearing a sentence as parts before interpreting it.

VISUAL AID

tokenization

FEEDBACK

Prompt tokens and generated response tokens both map to IDs, and IDs retrieve learned embedding vectors.

RELATIONSHIP

Prompt tokens and earlier response tokens in the input context become token IDs, embeddings, tensors, and temporary hidden states.

BRAIN LIMIT

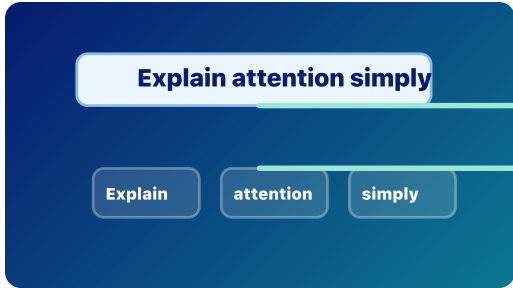
The parts are tokenizer chunks, not human phonemes, concepts, or meanings.

CURRENT EXERCISE

prompt-or-response-label: Prompt or Response?

REWRITE NOTES

Risk: Tokens may be mistaken for whole words or human concepts. **Missing:** Show that token boundaries can be word pieces, punctuation, or spaces, not necessarily words.



Text to Tokens
Text is split into model-readable chunks before embedding lookup.

Rubric Scores				
Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 5/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5
token	prompt tokens	response tokens	token-id	embedding

Token IDs

Chunks become lookup numbers

Stage: prompt processing

Priority: low

Rubric avg: 4.1/5

DEFINITION

A token ID is the number assigned to a token so the model can look up its learned starting vector.

METAPHOR

A library call number that points to the right shelf in the embedding table.

PROMPT/RESPONSE NOTE

This card should stay explicit that prompt tokens and response-so-far tokens enter the same current context before the next token is chosen.

CHECKPOINT

Why does the model need token IDs? Correct: To look up embeddings. Incorrect: To store a permanent memory; To skip tensors.

ILLUSTRATION NEEDED

Token chunks mapped to numeric IDs, then to rows in an embedding table.

CURRENT EXPLANATION

Where: It sits between tokenization and the embedding table. **Why:** The ID is the bridge from text chunks to the learned numerical space the model uses.

BRAIN METAPHOR

The lookup step is faintly like recognizing a familiar symbol.

VISUAL AID

token-ids

FEEDBACK

Token IDs connect text chunks to learned embedding vectors.

RELATIONSHIP

Prompt tokens and already-generated response tokens use token IDs when they are part of the current input context.

BRAIN LIMIT

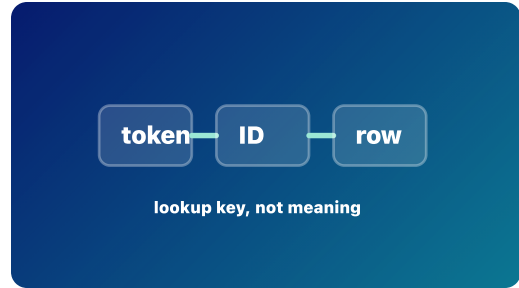
The number itself is not understanding; it is a key into a learned table.

CURRENT EXERCISE

prompt-or-response-label: Prompt or Response?

REWRITE NOTES

Risk: The ID can be mistaken for meaning rather than a lookup key. **Missing:** Make the tokenizer and embedding table two separate pieces in the diagram.



Token IDs
Each token gets a lookup number that points to an embedding row.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

token-id

prompt tokens

response tokens

embedding

Token IDs look up learned vectors

Stage: prompt processing Priority: low Rubric avg: 4.3/5

DEFINITION
An embedding is a token ID's learned starting vector.

METAPHOR
A token gets a starting address in a giant meaning cloud.

PROMPT/RESPONSE NOTE
This card should stay explicit that prompt tokens and response-so-far tokens enter the same current context before the next token is chosen.

CHECKPOINT
What is an embedding in this app? Correct: A learned starting vector. Incorrect: A finished sentence; A human memory.

ILLUSTRATION NEEDED
Lookup table pulling a learned starting vector for one token ID.

CURRENT EXPLANATION
Where: It is retrieved from the embedding table at the start of model processing. **Why:** Embeddings give each token a learned numerical starting point before context reshapes it.

BRAIN METAPHOR
It loosely resembles a starting association a person might have with a word.

VISUAL AID
embeddings

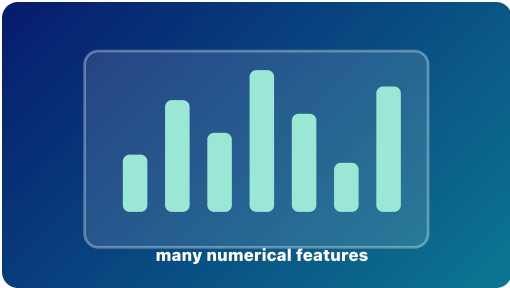
FEEDBACK
A token ID retrieves an embedding vector before context reshapes it into hidden states.

RELATIONSHIP
Prompt tokens and already-generated response tokens are looked up as embeddings when they are inside the current context.

BRAIN LIMIT
An embedding is not a definition or memory; it is a learned vector row.

CURRENT EXERCISE
prompt-or-response-label: Prompt or Response?

REWRITE NOTES
Risk: Embeddings can be confused with definitions, memories, or hidden states. **Missing:** Contrast embedding as starting vector with hidden state as later context-shaped vector.



Embedding Lookup
A token ID retrieves a learned starting vector.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

embedding input context token-id vector hidden state

Vectors

Lists of numbers carrying features

Stage: architecture

Priority: high

Rubric avg: 3.8/5

DEFINITION

A vector is a list of numbers that represents learned features of something, such as a token.

METAPHOR

A coordinate in a many-dimensional map.

PROMPT/RESPONSE NOTE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

CHECKPOINT

Which statement is most accurate? Correct: Embeddings and hidden states are vectors. Incorrect: Vectors are whole paragraphs; Vectors are permanent chats.

ILLUSTRATION NEEDED

Many-dimensional feature bars with a warning that labels are simplified.

CURRENT EXPLANATION

Where: Embeddings and hidden states are both vectors at different moments in the model path. **Why:** Vectors let the model compute with many fuzzy features at once instead of using one plain-English label.

BRAIN METAPHOR

It can resemble a bundle of associations around a word.

VISUAL AID

vectors

FEEDBACK

Vectors are lists of numbers. Embeddings are starting vectors; hidden states are temporary context-shaped vectors.

RELATIONSHIP

An embedding vector starts the token; hidden-state vectors are later context-shaped versions of that token.

BRAIN LIMIT

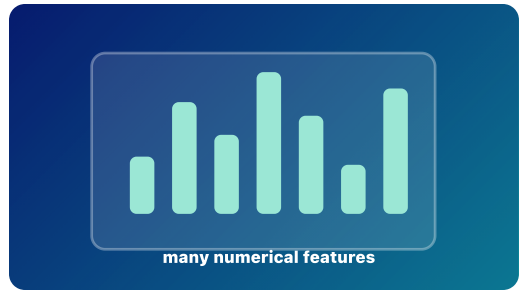
The vector is not a conscious concept; it is a numerical representation useful for computation.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Feature dimensions may be treated as literal labeled meanings rather than learned numerical patterns. **Missing:** Explain that dimensions are learned and distributed; one feature is rarely a neat English label.



Feature Vector

A vector is a list of numerical features, not a sentence.

Rubric Scores

Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 3/5	Exercise 2/5	Mobile 4/5

vector

embedding

hidden state

feature

Tensors

Organized blocks of vectors

Stage: architecture

Priority: high

Rubric avg: 3.7/5

DEFINITION

A tensor is an organized block of numbers, often holding many token vectors at once.

METAPHOR

A stack of spreadsheets carrying token features through the model.

PROMPT/RESPONSE NOTE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

CHECKPOINT

Which is the better definition of tensor here? Correct: An organized block of numbers. Incorrect: A word in a dictionary; A rule written by a programmer.

ILLUSTRATION NEEDED

Mobile-readable tensor block with token-position and feature axes.

CURRENT EXPLANATION

Where: Tensors carry the whole current context through transformer layers. **Why:** A tensor keeps the model's data organized by positions and features, so many tokens can be processed together.

BRAIN METAPHOR

It loosely resembles organizing many signals at once.

VISUAL AID

tensors

FEEDBACK

In transformers, tensors carry batches of token vectors through layers.

RELATIONSHIP

Embeddings from the current context become tensors so the transformer can process prompt tokens and earlier response tokens together.

BRAIN LIMIT

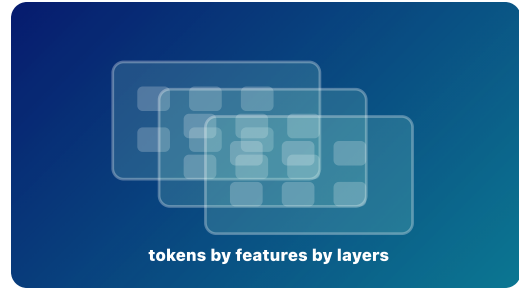
The tensor is not a mental image or thought; it is a shaped numerical array.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Tensor language can become jargon unless shapes and positions are made tangible. **Missing:** Show tensor axes: token position by feature, optionally across batches or heads only as a footnote.



Tensor Block
Tensors organize many token vectors for layer-by-layer processing.

Rubric Scores

Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5

tensor

vector

input context

layer

Attention

Weighted relevance between positions

Stage: architecture

Priority: high

Rubric avg: 4.4/5

DEFINITION

Attention assigns weighted relevance between token positions.

METAPHOR

Spotlights showing which earlier tokens are useful right now.

PROMPT/RESPONSE NOTE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

CHECKPOINT

Which tokens can attention look across during inference? Correct: Tokens currently in the context window. Incorrect: All text the model has ever seen; Only the newest generated token.

ILLUSTRATION NEEDED

Token nodes with weighted relevance arcs and a plain-language caption.

CURRENT EXPLANATION

Where: It happens inside transformer layers during a forward pass. **Why:** Attention lets one token position use information from other visible token positions in the current context.

BRAIN METAPHOR

The word attention is useful because some parts of the context matter more than others.

VISUAL AID

attention

FEEDBACK

Attention operates over the current context. It does not browse all training data.

RELATIONSHIP

During inference, attention operates over token positions in the current context, including prompt tokens and earlier generated response tokens.

BRAIN LIMIT

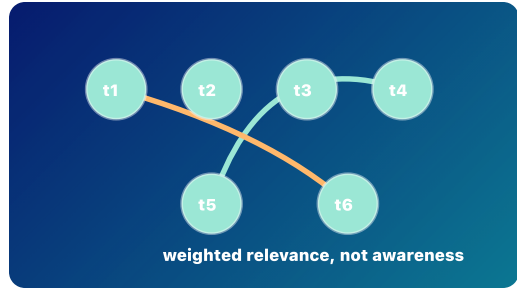
It is not human attention, awareness, desire, or focus. It is weighted relevance in a computation.

CURRENT EXERCISE

attention=relevance: Attention Is Relevance

REWRITE NOTES

Risk: The term attention invites human-focus and awareness metaphors. **Missing:** Show attention weights as arrows between positions, not as a mind selecting ideas.



Attention Weave
Attention is weighted relevance between token positions.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 5/5	Visual 4/5	Exercise 4/5	Mobile 4/5

attention

input context

prompt tokens

response tokens

hidden state

layer

Feature reshaping per token

Stage: architecture

Priority: high

Rubric avg: 4.3/5

DEFINITION

An MLP is a feed-forward network that reshapes each token position's feature vector.

METAPHOR

A forge that bends each token vector into a more useful shape.

PROMPT/RESPONSE NOTE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

CHECKPOINT

What does the MLP mainly do here? Correct: Reshapes per-token features. Incorrect: Reads private files; Chooses the final answer directly.

ILLUSTRATION NEEDED

Per-token vector forge or feature mixer after attention.

CURRENT EXPLANATION

Where: It appears inside transformer layers, usually after attention has mixed information across positions.
Why: The MLP helps turn mixed context into more useful per-token features for later layers.

BRAIN METAPHOR

It is somewhat like a small processing step after noticing relevant context.

VISUAL AID

mlp

FEEDBACK

MLPs transform token feature vectors after attention has mixed contextual information.

RELATIONSHIP

During a forward pass over the current context, attention mixes information across positions and the MLP reshapes features within each position.

BRAIN LIMIT

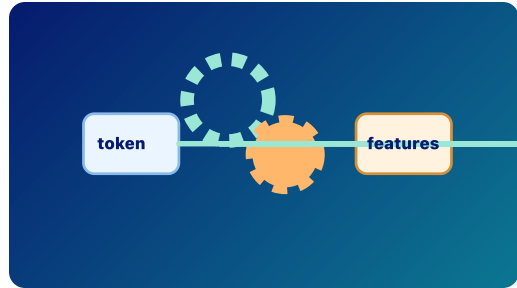
The MLP is not a neuron with feelings or beliefs; it is a learned function applied to vectors.

CURRENT EXERCISE

mlp-feature-reshape: MLP Feature Reshape

REWRITE NOTES

Risk: The MLP can blur into attention unless cross-position mixing and per-position reshaping are contrasted. **Missing:** Put MLP beside attention in the same layer so their complementary roles are visible.



MLP Reshape
The MLP reshapes each token position feature vector.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 4/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

MLP

input context

hidden state

attention

Layers

Repeated refinement

Stage: architecture

Priority: medium

Rubric avg: 3.9/5

DEFINITION

A transformer layer is a repeated block that usually combines attention, MLP feature reshaping, residual paths, and normalization.

METAPHOR

A series of editing passes where each pass can revise but also preserve the draft.

PROMPT/RESPONSE NOTE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

CHECKPOINT

What do layers repeatedly refine? Correct: Hidden states. Incorrect: Permanent user memory; The app interface.

CURRENT EXPLANATION

Where: Layers are stacked inside the model between embeddings and next-token scores. **Why:** Repeated layers let simple starting vectors become richer, context-shaped hidden states.

BRAIN METAPHOR

The stack can feel like stages of processing.

VISUAL AID

Layers

FEEDBACK

Layers update temporary internal vectors during inference.

RELATIONSHIP

Each layer refines hidden states while residual paths help useful information carry forward.

BRAIN LIMIT

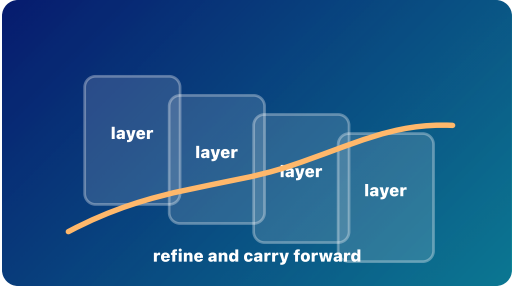
The layers do not form a human-like chain of thought; they update numerical states.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Layer stacks can be mistaken for a human chain of thought. **Missing:** Add residual paths and normalization as optional advanced labels or a split target lesson.



Layer Stack
Repeated blocks refine hidden states while carrying useful signal forward.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5

layer

attention

MLP

hidden state

Hidden States

Temporary contextual vectors

Stage: inference

Priority: high

Rubric avg: 4.4/5

DEFINITION

A hidden state is the model's temporary internal context-shaped vector for a token at a given layer.

METAPHOR

A scratchpad of numbers that changes while the prompt is being processed.

PROMPT/RESPONSE NOTE

This is the live prompt-to-response path: fixed weights process current context and generate response tokens.

CHECKPOINT

Why is it called hidden state? Correct: It is internal numbers, not visible text. Incorrect: It is encrypted English; it is a secret memory file.

ILLUSTRATION NEEDED

Same token glowing differently across contexts and layers.

CURRENT EXPLANATION

Where: Hidden states appear during the forward pass after embeddings enter the layer stack. **Why:** They are where the prompt's context changes the token representation before the model scores possible next tokens.

BRAIN METAPHOR

Hidden state can resemble temporary working memory.

VISUAL AID

hidden-states

FEEDBACK

Hidden means inside the model's working representation.

RELATIONSHIP

Hidden states are created while processing the current context during a forward pass; they are not the visible response.

BRAIN LIMIT

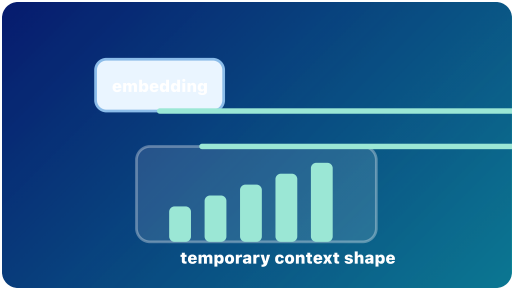
It is not permanent memory, consciousness, or private English text; it is a temporary vector.

CURRENT EXERCISE

mlp-feature-reshape: MLP Feature Reshape

REWRITE NOTES

Risk: Hidden states can be mistaken for secret English text or permanent memory. **Missing:** Show hidden state changing layer by layer for the same token in two contexts.



Temporary Hidden State
Hidden states are context-shaped vectors created during a forward pass.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

hidden state

input context

forward pass

embedding

layer

inference

Raw next-token scores

Stage: response generation

Priority: medium

Rubric avg: 4.4/5

DEFINITION

Logits are raw scores for the next token before they become probabilities.

METAPHOR

A scoreboard before the points are converted into odds.

PROMPT/RESPONSE NOTE

This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

CHECKPOINT

Where do probabilities come from in this path? Correct: Softmax converts logits. Incorrect: Training rewrites the weights; The model asks a database.

ILLUSTRATION NEEDED

Raw candidate scoreboard before softmax.

CURRENT EXPLANATION

Where: They appear near the end of a forward pass, after the final hidden state is projected toward the vocabulary. **Why:** Logits show that the model has not chosen a token yet; it has scored candidates.

BRAIN METAPHOR

It can feel like considering options.

VISUAL AID

logits

FEEDBACK

Logits are raw scores; softmax turns them into probabilities.

RELATIONSHIP

A forward pass over the current context produces logits for one next generated token.

BRAIN LIMIT

The model is not weighing options with intent; logits are raw numerical scores.

CURRENT EXERCISE

softmax-funnel: Softmax Funnel

REWRITE NOTES

Risk: Raw scores can be confused with probabilities or truth confidence. **Missing:** Make vocabulary projection explicit: final hidden state to one score per candidate token.



Raw Scores

Logits are raw next-token scores before probabilities.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

logits

generated token

forward pass

hidden state

softmax

Softmax

Scores become probabilities

Stage: response generation

Priority: medium

Rubric avg: 4.4/5

DEFINITION

Softmax turns next-token raw scores into probabilities that sum to one.

METAPHOR

A funnel turning scoreboard points into odds.

PROMPT/RESPONSE NOTE

This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

CHECKPOINT

What does softmax produce? Correct: Next-token probabilities. Incorrect: Permanent memory; A whole essay.

ILLUSTRATION NEEDED

Funnel converting raw scores into probabilities that sum to one.

CURRENT EXPLANATION

Where: It sits between logits and sampling during a decoding step. **Why:** Softmax makes scores comparable as a probability cloud over candidate tokens.

BRAIN METAPHOR

It resembles turning vague preferences into relative chances.

VISUAL AID

softmax

FEEDBACK

Softmax turns raw next-token scores into a probability distribution.

RELATIONSHIP

Softmax converts logits into probabilities; sampling then chooses one token from those probabilities.

BRAIN LIMIT

The probabilities are mathematical outputs, not confidence, desire, or truth.

CURRENT EXERCISE

softmax-funnel: Softmax Funnel

REWRITE NOTES

Risk: Probabilities can be mistaken for correctness or factual confidence. **Missing:** Show that softmax normalizes scores into a distribution that sums to one.



Softmax Funnel
Softmax converts raw scores into probabilities that sum to one.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

softmax

logits

decoding step

sampling

probability

Sampling

One next token is chosen

Stage: response generation

Priority: medium

Rubric avg: 4.3/5

DEFINITION

Sampling chooses one next token for the response from the probability cloud.

METAPHOR

Choosing from a weighted bowl of possible next tokens.

PROMPT/RESPONSE NOTE

This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

CHECKPOINT

When the model chooses one token, what happens next? Correct: The token is appended and the model runs again. Incorrect: The model writes the whole response at once; The model permanently learns the token.

ILLUSTRATION NEEDED

Weighted vocabulary cloud with one selected token.

CURRENT EXPLANATION

Where: It happens after softmax in the decoding step.
Why: Sampling explains why the same prompt can sometimes produce different good responses.

BRAIN METAPHOR

It can resemble choosing a word while speaking.

VISUAL AID

sampling

FEEDBACK

The selected response token becomes part of the context for the next forward pass.

RELATIONSHIP

Temperature, top-k, and top-p shape the candidate pool before the chosen response token is appended to the context.

BRAIN LIMIT

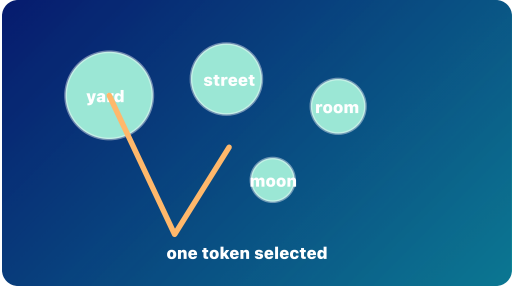
The model is not choosing with intention; a decoding rule selects from probabilities.

CURRENT EXERCISE

pick-next-token: Pick the Next Token

REWRITE NOTES

Risk: Randomness can sound arbitrary unless tied to weighted probabilities. **Missing:** Add temperature/top-p as knobs that shape selection without overexplaining math.



one token selected

Sample One Token

Sampling chooses one token from the probability cloud.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

sampling

generated token

decoding step

response tokens

temperature

top-k

top-p

Stage: response generation

Priority: low

Rubric avg: 4.4/5

DEFINITION

Autoregression means response generation by repeated next-token prediction: choose one token, append it, and run again.

METAPHOR

A train adding one car at a time.

PROMPT/RESPONSE NOTE

This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

CHECKPOINT

How does an LLM normally generate text? Correct: One token at a time. Incorrect: All words at once; By rewriting its weights.

CURRENT EXPLANATION

Where: It is the loop that builds text during inference.
Why: Autoregression explains why earlier generated tokens become part of what the model can use next.

BRAIN METAPHOR

It resembles continuing a thought step by step.

VISUAL AID

autoregression

FEEDBACK

LLM generation is autoregressive: next token, append, repeat.

RELATIONSHIP

Each generated response token becomes part of the input context for the next decoding step.

BRAIN LIMIT

The model is not holding a plan in mind; it repeatedly predicts the next token from visible context.

CURRENT EXERCISE

prompt-or-response-label: Prompt or Response?

REWRITE NOTES

Risk: Learners may still imagine the model drafting a full paragraph internally before showing it. **Missing:** Use one short dog/cat sentence to show next token, append, repeat.

next

append

run again

next token, append, repeat

Append and Repeat

The chosen token is appended, then the model runs again.

Rubric Scores				
Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

autoregression

completion

generated token

response tokens

sampling

inference

Context Window

Temporary working context

Stage: current context Priority: high Rubric avg: 4.5/5

DEFINITION
A context window is the limited amount of tokens or media the model can currently use.

METAPHOR
A moving spotlight over a growing train.

PROMPT/RESPONSE NOTE
This card should separate the temporary visible context from durable memory or weight updates.

CHECKPOINT
What can happen when the context window is full? Correct: Older tokens may be truncated. Incorrect: The model permanently learns them; Softmax disappears.

ILLUSTRATION NEEDED
Limited stack of cards where older cards fall out.

CURRENT EXPLANATION
Where: It surrounds the prompt, conversation history, retrieved material, and response-so-far that remain visible. **Why:** It explains why models can use current context but do not automatically form permanent memory from every chat.

BRAIN METAPHOR
It is loosely like working memory: what is currently available matters.

VISUAL AID
context-window

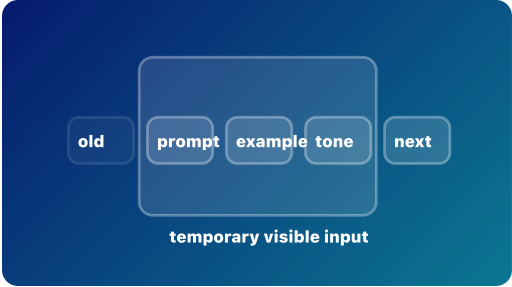
FEEDBACK
Context is temporary input, not durable training.

RELATIONSHIP
The model can only use tokens currently inside the context window, whether they came from the prompt, prior conversation, retrieved documents, or the response-so-far.

BRAIN LIMIT
It is not durable memory. What falls out cannot be directly attended to in the next pass.

CURRENT EXERCISE
context-window-fell-out: Context Window: What Fell Out?

REWRITE NOTES
Risk: Current context can be confused with saved memory or training data. **Missing:** Show what falls out and that retrieval adds text to context rather than training weights.



Temporary Context Window
Only visible context can influence the next token.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

context window input context prompt tokens response tokens rag autoregression inference

RAG and Retrieval

Retrieved text joins the current context

Stage: current context Priority: high Rubric avg: 4.6/5

DEFINITION

Retrieval-augmented generation, or RAG, retrieves outside information and places it into the model's context before generating a response.

METAPHOR

Open-book AI: the model gets notes before answering.

PROMPT/RESPONSE NOTE

Retrieved documents become prompt/context tokens. The answer is still generated as response tokens one at a time.

CHECKPOINT

What does RAG usually do? Correct: Retrieves outside information and places it into the current context. Incorrect: Permanently updates the model's weights; Makes the model conscious of documents; Guarantees that every answer is true.

ILLUSTRATION NEEDED

Three lanes: user question, retriever searches a document shelf, retrieved cards enter the context window, and the LLM generates response tokens.

CURRENT EXPLANATION

Where: It happens during inference, before or during response generation, and usually does not change model weights. **Why:** RAG helps learners see that the model is not magically knowing private documents; retrieved snippets become temporary context.

BRAIN METAPHOR

It is a little like looking something up in notes before answering a question.

VISUAL AID

rag-retrieval

FEEDBACK

RAG is retrieval plus context. It can improve grounding, but it is not training, fine-tuning, permanent memory, or a truth guarantee. Choice notes: Permanently updates the model's weights: Not quite. Unless a separate training process happens, RAG does not durably update model weights. Makes the model conscious of documents: Not quite. Retrieved documents become visible context; the model does not become aware of them. Guarantees that every answer is true: Not quite. RAG can reduce hallucinations, but retrieval can be poor and the model can still misuse context.

RELATIONSHIP

RAG depends on the context window: retrieved text becomes prompt/context tokens, then the model still uses attention, hidden states, logits, softmax, and sampling to generate response tokens.

BRAIN LIMIT

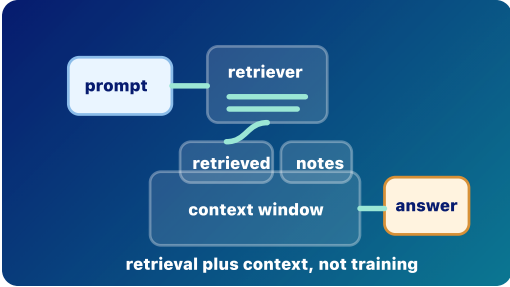
A human can judge sources, remember what was read, and reason about trust in richer ways. The model still generates likely response tokens from retrieved context and learned weights.

CURRENT EXERCISE

open-book-or-Learned: Open Book or Learned?

REWRITE NOTES

Risk: RAG can be mistaken for permanent learning, full file access, web search by itself, fine-tuning, or a guarantee that answers are true. **Missing:** Name the retrieval step, the context insertion step, and the fact that weights remain unchanged unless training also happens.



Open-Book Retrieval

Retrieved document cards enter the context before response tokens are generated.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 5/5	Exercise 4/5	Mobile 4/5

rag retrieval grounding context window input context prompt tokens response tokens inference fine-tuning training hallucination

Durable learning versus temporary steering

Stage: risk/policy

Priority: high

Rubric avg: 3.9/5

DEFINITION

AI systems can improve or behave differently through durable training, targeted fine-tuning, retrieval, or temporary in-context steering.

METAPHOR

Revising the textbook versus bringing notes into an open-book exam.

PROMPT/RESPONSE NOTE

This card uses the prompt/response path to separate practical risk from myths about secret learning or agency.

CHECKPOINT

Which process normally changes only the current context, not weights? Correct: Retrieval-augmented generation. Incorrect: Pretraining; Fine-tuning.

ILLUSTRATION NEEDED

Matrix of durable weight change, temporary context steering, and retrieval.

CURRENT EXPLANATION

Where: These processes happen at different times: before deployment, between deployments, or inside one current run. **Why:** Calling every behavior change learning creates confusion about privacy, memory, and institutional risk.

BRAIN METAPHOR

The word learning is useful when prior experience changes later behavior.

VISUAL AID

ai-learns

FEEDBACK

RAG places retrieved material into the current context. It does not normally train the base model.

RELATIONSHIP

Pretraining and fine-tuning can durably change weights; retrieval and examples in the prompt change only what the model can see during inference.

BRAIN LIMIT

In-context examples are not personal learning; they steer the current run without normally updating weights.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Calling every behavior change learning can blur training, retrieval, and in-context steering. **Missing:** Separate durable training, retrieval, temporary instructions, and deployed online learning in one table.



Learning Modes

Durable training, retrieval, and temporary steering change different things.

Rubric Scores

Accuracy 4/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 3/5

training

pretraining

fine-tuning

inference

rag

in-context learning

Diffusion vs Autoregression

Denoising is a different generation pattern

Stage: architecture

Priority: medium

Rubric avg: 3.6/5

DEFINITION

Diffusion generation usually starts from noise and denoises step by step.

METAPHOR

Writing the next word versus developing a photograph out of static.

PROMPT/RESPONSE NOTE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

CHECKPOINT

What does a diffusion model usually start with? Correct: Noise. Incorrect: A completed paragraph; A spreadsheet.

ILLUSTRATION NEEDED

Denoising sequence beside a next-token text loop.

CURRENT EXPLANATION

Where: Diffusion is common in image generation and other media workflows, not the standard loop for text LLM generation. **Why:** The comparison prevents learners from treating all generative AI as the same mechanism.

BRAIN METAPHOR

Both systems can feel imaginative because they produce new outputs.

VISUAL AID

diffusion

FEEDBACK

Diffusion models learn to denoise toward useful outputs.

RELATIONSHIP

Autoregressive LLMs build text token by token; diffusion models refine noise into an image or other output.

BRAIN LIMIT

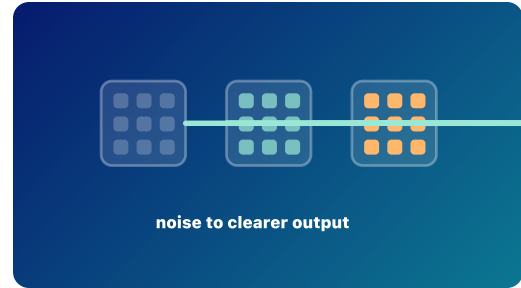
The mechanisms differ: diffusion denoises a representation; autoregressive text predicts and appends tokens.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Learners may overgeneralize diffusion mechanics to text LLM generation. **Missing:** Show a side-by-side: denoise image representation versus append text tokens.



Denoise, Not Append
Diffusion refines noise step by step instead of generating text token by token.

Rubric Scores				
Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 2/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5
diffusion	autoregression			

Multimodal AI

Multiple media types together

Stage: architecture

Priority: medium

Rubric avg: 3.7/5

DEFINITION

Multimodal AI can represent or process multiple media types, such as text, images, audio, or video.

METAPHOR

A transit hub where different kinds of information can transfer lines.

PROMPT/RESPONSE NOTE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

CHECKPOINT

What does multimodal mean? Correct: Multiple media types. Incorrect: A larger context window only; A model with feelings.

ILLUSTRATION NEEDED

Text, image, audio, and video paths meeting in a shared representation hub.

CURRENT EXPLANATION

Where: It appears when systems connect encoders, decoders, or shared representations across media types. **Why:** It explains how a system can read an image, answer in text, or combine speech and documents.

BRAIN METAPHOR

People combine sight, speech, and text naturally, so the metaphor helps.

VISUAL AID

multimodal

FEEDBACK

Multimodal systems work across media types, often through learned representations.

RELATIONSHIP

Different modalities may use connected encoders, shared vector spaces, or coordinated model components.

BRAIN LIMIT

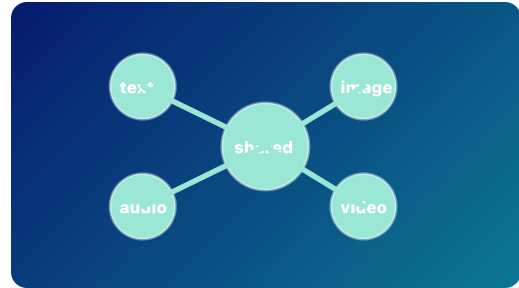
Machine modalities are engineered representations, not human sensory experience.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Human sensory metaphors can make engineered media representations sound more human-like than they are. **Missing:** Add an example of image input, text output, and the representation bridge between them.



Shared Media Hub
Different media types can connect through learned representations.

Rubric Scores				
Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5

multimodal

embedding

vector

Stage: risk/policy

Priority: high

Rubric avg: 4.3/5

DEFINITION

Risk literacy separates practical harms from magical stories about what models are.

CURRENT EXPLANATION

Where: It matters whenever people use AI in classrooms, research, administration, health, advising, or IT systems. **Why:** Clear mental models help institutions manage privacy, accuracy, bias, over-trust, and security without inventing magical threats.

RELATIONSHIP

Most real risks come from data, outputs, integrations, incentives, and human over-reliance; ordinary inference does not make the model secretly train itself.

METAPHOR

A campus safety map that marks real hazards clearly.

BRAIN METAPHOR

Brain metaphors can make risks feel urgent and memorable.

BRAIN LIMIT

Do not infer consciousness, intent, or secret self-training from fluent text. Keep the mechanism visible.

PROMPT/RESPONSE NOTE

This card uses the prompt/response path to separate practical risk from myths about secret learning or agency.

VISUAL AID

risk

CURRENT EXERCISE

open-book-or-learned: Open Book or Learned?

CHECKPOINT

Which is a real institutional risk? Correct: Uploading private data. Incorrect: The model becoming conscious in the chat; Softmax stealing files.

FEEDBACK

Risk literacy is part of model literacy.

REWRITE NOTES

Risk: Fluent outputs can invite myths about consciousness, intent, or secret self-training. **Missing:** Tie each risk to a mechanism: context exposure, tool access, hallucination, bias, or over-trust.



Risk or Myth
Clear mechanisms help separate practical risk from magical stories.

ILLUSTRATION NEEDED
Risk sorting cards linked to mechanisms rather than fear language.

Rubric Scores				
Accuracy 5/5	Placement 4/5	Relationship 5/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5
inference	training	context window	privacy	hallucination
		prompt-injection		